

Savitzky-Golay optimizer diagnostics

Objective: mean squared translational/angular acceleration residual; angular residual uses the reference inertia/mass metric

Tolerances: {'ftol': None, 'xtol': None, 'gtol': 1e-06, 'note': 'ftol and xtol are disabled by default in the SG estimator; gtol or max_nfev terminates the solve unless explicitly overridden'}

Selected strict-ZOH solve

status=1 success=True nfev=4 njev=4

message='gtol' termination condition is satisfied.

reported optimality=1.8594785267112002e-08

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initial={'cost': 1.4156315076964658, 'gradient_inf_norm': 0.05221419046103826,
'gradient_l2_norm': 0.06510024523463871, 'jacobian': {'condition_number_nonzero_subspace':
407.3398381326683, 'numerical_rank': 14, 'numerical_tolerance': 9.540614684264724e-11,
'singular_values': [119.28680937518072, 118.13898883626935, 102.64038502137774,
48.60576762874349, 44.725731233208535, 25.867104260760534, 12.987971803006689,
11.719762536981703, 11.325383250703641, 11.045781651493806, 9.878416397906813,
3.7982524361265533, 3.7926876963020293, 0.29284346437121545]}, 'residual_l2_norm':
1.682635734611901}
final={'cost': 1.4156311624620588, 'gradient_inf_norm': 1.8594785267112002e-08,
'gradient_l2_norm': 3.011991092196547e-08, 'jacobian': {'condition_number_nonzero_subspace':
406.9245366549322, 'numerical_rank': 14, 'numerical_tolerance': 9.540621366493406e-11,
'singular_values': [119.28689292343891, 118.13837775114501, 102.64030055744644,
48.60572321038924, 44.72568727550868, 25.867075635050803, 12.988033801746914,
11.719751814942125, 11.325338914731379, 11.04581132508363, 9.87842978820393,
3.7982364108291167, 3.7927000517437874, 0.29314254162214104]}, 'residual_l2_norm':
1.6826355294371143}
coordinate_step_l2_norm=0.0005127896673439322
cost_reduction=3.452344070620228e-07
gradient_tolerance_satisfied=True
finite_difference_max_relative_error=7.904933618886317e-12
finite_difference_directional_checks=[{'analytic_l2_norm': 98.8487221224677,
'difference_l2_norm': 7.813925866911972e-10, 'direction': 'negative_gradient',
'finite_difference_l2_norm': 98.84872212263856, 'relative_error': 7.904933618886317e-12,
'scheme': 'central', 'step': 6.0554544523933395e-06}]
```

All smoothstep and strict-ZOH stage details are in optimizer_diagnostics.json.